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USL Assignment 6

t-SNE – t-Distributed Stochastic Neighbor Embedding for Dimensionality Reduction

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TSNE THEORY

1. Introduction to t-SNE

1. Introduction to t-SNE

t-SNE (t-Distributed Stochastic Neighbor Embedding) is an **unsupervised machine learning technique** used for **dimensionality reduction and visualization of high-dimensional data**.

- It transforms complex high-dimensional data into a **low-dimensional space (2D or 3D)**
- Primarily used for **data visualization and pattern discovery**
- Unlike clustering algorithms such as K-Means, t-SNE does not group data but **reveals structure in data**

It is widely used in:

- Image data visualization
- Natural language processing
- Pattern recognition

2. Working of t-SNE Algorithm

t-SNE works by converting distances between data points into **probability distributions**.

Step-by-step process:

1. Compute similarity between points in **high-dimensional space**
2. Convert distances into probabilities Pij
3. Initialize points randomly in a **low-dimensional space**
4. Compute similarity Qij in low-dimensional space
5. Minimize the difference between Pij and Qij
6. Update positions using gradient descent

3. Probability Distribution of Neighbors

t-SNE defines similarity using probabilities:

- **High-dimensional space:**

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2)}$$

- Measures how similar two points are
- Nearby points → higher probability

- **Low-dimensional space:**

$$\mathbf{Q_{ij}} = \frac{(\mathbf{1} + ||\mathbf{y_i} - \mathbf{y_j}||^2)^{-1}}{\sum_{\mathbf{k} \neq \mathbf{i}}(\mathbf{1} + ||\mathbf{y_i} - \mathbf{y_k}||^2)^{-1}}$$

- Uses **Student's t-distribution**
- Helps reduce the **crowding problem**

4. KL Divergence (Cost Function)

t-SNE minimizes the difference between the two distributions using:

$$\boldsymbol{KL(P||Q) = \sum_i \sum_j P_{ij} \log \frac{P_{ij}}{Q_{ij}}}$$

Interpretation:

- Measures how different the two distributions are

- Lower value → better representation

5. Key Concept: Local Structure Preservation

t-SNE focuses on:

- Keeping **similar points close together**
- Preserving **local neighborhoods**

6. Parameters of t-SNE

🔹 **Perplexity**

- Controls number of neighbors considered
- Typical range: 5–50

🔹 **Learning Rate**

- Controls speed of optimization

🔹 **Number of Components**

- 2 → 2D visualization
- 3 → 3D visualization

8. Advantages of t-SNE

- Excellent for visualization of high-dimensional data
- Reveals hidden patterns and clusters
- Preserves local structure effectively

9. Limitations of t-SNE

- Computationally expensive (slow for large datasets)
- Not suitable for predictive modeling
- Results may vary due to randomness
- Does not preserve global structure